

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2082 Bhadra

Exam.	Regular	
Level	BE	Full Marks 80
Programme	BEI	Pass Marks 32
Year / Part	IV / I	Time 3 hrs.

Subject: - Wireless Communication (EX 715)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Discuss the basic trends in wireless communication beyond 3G, focusing on key technological changes. [4]
2. a) Explain two schemes for prioritizing handoff. Define terms grade of service, traffic intensity, average holding time and block call. [3+4]
- b) The cellular system has 32 cells, each cell has a 1.6 km radius and the system reuse factor is 7. The system is to support 336 traffic channels in total. Determine the total geographical area covered, the number of traffic channels per cell and the total number of simultaneous calls supported by this system. [4]
3. a) Define the significance of term frequency dispersion in wireless communication. Derive the relation for doppler spread. [2+4]
- b) Find the mean path loss using Okumura's Model for $d = 50$ km, $h_{te} = 100$ m, $h_{re} = 10$ m in a suburban environment. If the BS transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find EIRP in dBm and the power at the receiver where gain of receiving antenna is 10 dB. (Given: $A_{mu}(f,d) = 43$ dB, $G_{Area}(f) = 9$ dB) [5]
- c) Define the terms coherence bandwidth and coherence time. What are the factors that influenced in indoor propagation model? Explain. [2+5]
4. Define spread spectrum modulation. Explain direct sequence spread spectrum with the block diagrams of transmitter and receiver. [2+6]
5. Describe a signal processing operation that minimizes the effects of ISI. What do you understand by RAKE receiver? Explain the structure of linear transversal equalizer with appropriate diagram. [2+1+4]
6. a) Define turbo coding. Construct 8 by 8 Hadamard code. [2+3]
- b) What are different characteristics of speech signals? How they are used in designing of coders? [4]
7. a) Explain the implementation of CDMA with its encoding part and decoding part with relevant example. [7]
- b) Explain the block diagram of OFDM transmitting side. Also mention its advantage and disadvantage. [4+4]
8. Explain the components of Network Switching Subsystem in GSM architecture. Illustrate GSM Frame hierarchy in GSM system. [4+4]

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Year / Part	IV / I	Time	3 hrs.

Subject: - Wireless Communication (EX 715)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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1. Briefly explain the evolution of different generations of cellular systems. [6]
2. What do you mean by frequency reuse? Explain various channel assignment strategies used in mobile communication. [2+4]
3. Consider a single high-power transmitter that can support 40 voice channels over an area of 140 km^2 with the available spectrum. If this area is equally divided into seven smaller cells, each supported by lower power transmitters so that each cell supports 30% of the total voice channels, then determine
 - i) Coverage area of each cell [1]
 - ii) Total number of voice channels available in a cellular system compared to a non-cellular system. [3]
4. Find the median path loss using Okumura's model for $d = 50 \text{ km}$, $h_{te} = 100 \text{ m}$, $h_{re} = 10 \text{ m}$ in a suburban environment. If the base station transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find the power at the receiver (assume a unity gain receiving antenna). [Assume $A_{mu} = 43 \text{ dB}$, $G_{AREA} = 9 \text{ dB}$] [8]
5. Define the terms coherence bandwidth and coherence time explaining their significance in mobile radio propagation. Explain fading with its various types. [3+5]
6. With block diagrams, describe the working principle of OFDM transmitter and receiver. [8]
7. Describe a signal processing operation that minimizes the effects of ISI. What do you understand by RAKE receiver? Explain the working of a M branch RAKE receiver. [2+2+4]
8. What are the characteristics of speech signals? Explain the basic working principle of Formant and Channel vocoder. [4+4]
9. What are the advantages of CDMA over TDMA? Explain the working of the Frequency Hopping Spread Spectrum (FHSS) multiple access technique with an appropriate block diagram. Highlight different types of FHSS. [4+6+2]
10. List the various types of control channels available in GSM with their purposes. [2+4]
11. Describe forward CDMA (IS-95) channel with suitable figure. [6]

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- List major benefits of wireless communication. Give a brief description of the 4G wireless communication system. [2+4]
- Explain the difference between co-channel and adjacent channel interference. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + ij + j^2$ is the cluster size. [3+3]
- A telephone network company needs to expand its capacity based on demand on a city. A group of engineers was selected to find the solution. Among the solution sectoring and cell splitting were major technique for expansion purpose. Being a cellular planning engineer, which option do you think is best and why? [4]
- Derive an expression for path difference and phase difference between line of sight (LOS) path and ground reflected (GR) path in two-ray model using simple geometrical relations. [6]
- Compare Okumura model with Hata model. Calculate mean excess delay, rms delay spread and maximum excess delay for the multipath profile $Pr(\tau)$ given as: $Pr(\tau) = (-20dB, -10dB, -10dB, 0dB)$ for $\tau = (0, 1, 2, 5)$ second. Also estimate 50% coherence bandwidth. [4+6]
- Explain the transmission and detection process of QPSK modulation technique with appropriate figure and constellation diagram. [8]
- Elaborate the working principle behind Interleaving with an appropriate diagram. Explain the working of Maximum Likelihood Sequence Estimation (MLSE). [4+4]
- Describe vocoders with block diagram. Briefly explain different kind of vocoders. [3+5]
- Explain any two types of hybrid spread spectrum multiple access techniques along with their advantage and disadvantage. Describe the operation of FHMA. What are the types of FHMA? [6+4+2]
- Explain the operation of different control channels used in the GSM system. [8]
- Describe regulatory issues related to licensing and spectrum allocation. [4]

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1. What do you mean by duplexing? Explain its different types. [2+4]
2. a) For a cluster $N = 7$ system with a blocking probability of 1% and an average call length of two minutes, find the traffic capacity loss for a blocked calls cleared system due to trunking for 57 channels when going from omnidirectional antennas to 60° sector antennas. The average request rate per user is 1 call per hour. [2+2+2]

GOS/ Channel	0.5%	1%	2%
1	.005	.0101	.0204
9	3.333	3.783	4.345
19	10.33	11.23	12.33
54	39.47	41.51	44.00
57	42.1	44.2	46.8

- b) What is Hand-off? Explain proper and improper HO with necessary drawing. [1+3]
3. a) What is Doppler spread? Derive the expression for Doppler shift. [2+4]
- b) If a transmitter produces 50 watts of power, express the transmit power in units of (i) dBm, and (ii) dBW. If 50 watts is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km)? Assume unity gain for the receiver antenna. [1+1+2+2]
- c) Define fading. Show its different types based on delay spread and Doppler spread. [1+3]
4. Explain the transmission and detection process of GMSK modulation technique with appropriate figure and constellation diagram. [8]
5. Define Interleaving. Explain how equalization offsets the ISI introduced by the multipath time-dispersive channel using an example of an adaptive equalizer. [2+6]
6. a) What are different characteristics of speech signals? How they are used in designing of codes? [4]
- b) With the help of a block diagram, explain the operation of a vocoder. [6]
7. a) Explain the working of the Direct Sequence Spread Spectrum (DSSS) technique with an appropriate block diagram. [6]
- b) Compare and contrast between TDMA and FDMA. [4]
8. a) Explain GSM signal processing with necessary figures. [8]
- b) Draw Reverse CDMA (IS-95) channel. [4]

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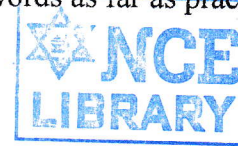


1. a) Compare and contrast the second, third and fourth generation of mobile communication standards in terms of technology advancement. [4]
- b) Define forward and reverse channel. [2]
2. a) A wireless communication transmitter has an output power of 165 watts at a carrier frequency of 325 MHz. It is connected to an antenna with a gain of 12 dBi. The receiving antenna is 15 km away and has a gain of 6 dBi. Calculate the power delivered to the receiver, considering free-space propagation. (Assume that there are no other losses or mismatches in the system.) [6]
- b) Derive the formula for co-channel reuse ratio: $Q = \sqrt{3N}$. [4]
3. a) Define path loss. Explain the parameters of mobile multipath channels used to classify various types of fading. [2+4]
- b) Determine the mean path loss using Okumura's model for $d = 50$ km, $h_{te} = 100$ m, $h_{re} = 10$ m in a suburban environment. If the base station transmitter radiates an EIRP of 1kW at a carrier frequency of 900 MHz. Find EIRP (dBm) and the power at the receiver where the gain at receiving antenna is 10 dB. The medium attenuation factor is 43 dB and the gain due to area is 9 dB for a 50 km distance and 900 MHz frequency of operation. [4+2+2]
- c) What do you mean by Ericsson Multiple Breakpoint model? [2]
4. Explain the operation of Orthogonal Frequency Division Multiplexing (OFDM) with an appropriate block diagram of the transmitter and receiver. [8]
5. a) Why is there a need to implement diversity? Describe the Maximum ratio combining space diversity technique. [2+2]
- b) Why interleaving is needed in wireless communication? Explain its working in brief. [4]
6. a) Explain sub-band coding with transmitter and receiver. [6]
- b) Construct Hadamard matrix for H_8 . [4]
7. a) Explain the working of FHMA technique with an appropriate block diagram. [6]
- b) Explain any two hybrid multiple access techniques. [4]
8. a) Explain the operation of the Broadcast control channel (BCH) and Common control channel (CCCH) in GSM. [4+4]
- b) Draw forward CDMA (IS-95) channel. [4]

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1. Compare and contrast third and fourth generation of mobile communication standards in terms of technology advancement. [4]
2. a) An urban area has a population of two million residents. System A has 394 cells with 19 channels each. Find the number of users that can be supported at 2% blocking if each user average two calls per hour at average call duration of three minutes. Assuming that trunk systems are operated at maximum capacity, compute the percentage market penetration of cellular provider. [5]
b) Why we need microcell zone concept? Explain microcell zone concept in brief. [3]
3. a) Describe briefly the types of small scale fading in radio propagation. Explain the factors which influences small scale fading. [4+4]
b) Assume free space propagation, a receiver is located 10 km away from a 50 w transmitter. The carrier frequency is 900 MHz. antenna gain at transmitter and receiver are 1 and 2 respectively, find: [8]
 - i) The power received at the receiver
 - ii) The magnitude of E-field at the receiver antenna
 - iii) The power flux density
 - iv) The rms voltage applied to the receiver input.

The receiver antenna has a purely real impedance of 50Ω and is matched to the receiver.
4. How can Minimum Shift Keying (MSK) be defined as a special type of continuous phase FSK and Orthogonal Quadrature Phase Shift Keying (OQPSK)? Compare OQPSK, MSK and GMSK in terms of their bandwidth efficiency and power efficiency. Which of them are suitable for modulation in GSM and Why? [3+3+2]
5. Why Equalization and diversity are needed in wireless communication systems? List out different diversity reception methods used for space diversity. Explain any one of them. [3+3+2]
6. What are the characteristics of speech signal? Explain about Linear Predictive Coders (LPC) with neat block diagram. [2+6]
7. a) Explain the non-linear effect in FDMA. If the total spectrum allocation is 25 MHz, the guard band allocated at the edge of the spectrum is 100 KHz, and the channel bandwidth is 200 KHz. Find the number of channels available in an FDMA system. [3+2]
b) Explain the Frequency Hopped Multiple Access (FHMA) technique using its transmitter and receiver blocks. [7]
8. Explain the operation of each component in GSM architecture. [8]
9. Write short notes on: (Any Two) [2×4]
 - a) CDMA design considerations
 - b) Regulatory issues in wireless communication
 - c) Interference in wireless/mobile communication

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Year / Part	IV / I	Time	3 hrs.

Subject: - Wireless Communication (EX 715)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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1. Define forward and reverse channel. How 5G will be different from 4G mobile communication. [2+2]
2. Define Grade of Service (GoS) and explain how can it be measured in a 'blocked call cleared' type of trunking system. A cellular service provider decides to use a digital TDMA scheme that can tolerate a signal-to-interference ratio of 15 dB in the worst case. The mobile radio channel provided a propagation path loss exponent of $n = 3$. Find the optimal value of N for (a) Omni-directional antennas, (b) 1200 sectoring, and (c) 60° sectoring. Comment on your results. [3+5]
3. a) What do you mean by diffraction in radio wave propagation? Derive an expression for phase difference in Fresnel Zone Geometry model of direction. [2+6]
- b) In mobile propagation in a cellular system, find the correction factor and pathloss for a medium size city assuming carrier frequency as 950 MHz, height of transmitting antenna at base station is 45 m, propagation distance between antennas is 10 km and height of receiving antenna in mobile station is 5 m. Compute free space pathloss and compare it with Hata pathloss. [4+4]
4. Mention advantages of digital modulation over analog modulation. Explain the transmission and detection process of GMSK modulation scheme with block diagram and constellation diagram. [2+6]
5. Determine the optimal solution of the weights using MSE algorithm with its appropriate diagram and derivation. [8]
6. What are the characteristics of speech signal? Explain the operation of linear predictive coder. [3+5]
7. a) What do you understand by CDMA? Mention its characteristics, advantages and limitations. [2+6]
- b) Explain working of frequency Hopping Multiple Access with an example. [4]
8. Draw and explain the global System for Mobile (GSM) architecture in brief. [8]
9. Write short notes on: (Any Two) [2×4]
 - a) Practical handoff considerations
 - b) RAKE receiver
 - c) Spectrum regulations

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2080 Chaitra

Exam.	Regular		
Level	BE	Full Marks	80
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Year / Part	IV / II	Time	3 hrs.

Subject: - Wireless Communication (EX 751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
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1. Define forward and reverse channel. List out the features of 5G. [2+2]
2. a) Explain following terminologies [2.5+2.5]
 - (i) Co-channel interference
 - (ii) Adjacent interference
- b) What is cell footprint? Explain Interference tier in brief. [1+4]
3. What are the necessary conditions for Okumara and Hata Model? [2+2]
4. a) What is frequency selective fading? Explain Rayleigh distribution in brief. [2+4]
- b) Define Doppler's effect in Small Scale Fading. (i) Given a cordless phone operating at 52000 kHz frequency which has a range of 50m. Determine the free space path loss. [2+4]
5. What is MSK Modulation? Explain direct sequence spread spectrum with its necessary relevant diagram. [2+5]
6. Determine the optimal solution of the weights using MSE algorithm with its appropriate diagram and derivation. [7]
7. Explain the characteristics of Speech. Explain 7-bit hamming code method with relevant example. [3+5]
8. a) List out important features of CDMA in wireless communication. Illustrate hybrid spread spectrum techniques in brief. [2+6]
- b) In the N-TDMA system, it uses one way bandwidth of 25 MHz for the forward (or reverse) channel. The system bandwidth is divided into radio channels of 30 kHz. Guard bands with $B_g = 20\text{kHz}$ are used. Calculate number of simultaneously transmitting users that can be accommodated in each cluster is. [4]
- c) Explain OFDM receiver in brief. [4]
9. a) Illustrate Forward CDMA (IS -95) Channel in brief. [6]
- b) What is a significance of spectrum Management? What are the functions of the regulatory. [2+4]

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- ✓ Necessary table is attached herewith.



1. Define forward and reverse channel. How 5G will be different from 4G mobile communication. [2+2]
2. a) An urban area has a population of two million residents. System A has 394 cells with 19 channels each. Find the number of users that can be supported at 2% blocking if each user averages two calls per hour at average call duration of three minutes. Assuming that trunk systems are operated at maximum capacity, compute the percentage market penetration of cellular provider. [5]
 - b) Why we need microcell zone concept? Explain microcell zone concept in brief. [1+3]
3. What are the necessary conditions for Okumara and Hata Model? [2+2]
4. a) What is frequency selective fading? Explain the relationship between coherence bandwidth and delay spread in time domain. [2+7]
 - b) Explain basic propagation models. Compute the far field distance for an antenna with maximum dimension of 1 m and operating frequency of 900 MHz. [3+3]
5. Define spread spectrum? Explain frequency hopping spread spectrum with it's necessary relevant diagram. [2+5]
6. Determine the optimal solution of the weights using MSE algorithm with its appropriate diagram and derivation. [7]
7. Write the steps to satisfy condition for Hadamard code. Construct 8 by 8 Hadamard code that satisfies above conditions. [3+5]
8. a) Illustrate Encoding and decoding mechanism of CDMA system in brief. [8]
 - b) Calculate the number of radio channels available in the FDMA system with following data. A US analog mobile phone system is allocated 12.8 MHz fir every simplex band. The total spectrum allocated is 12.8 MHz, the guard bandwidth is 10 KHz and the channel bandwidth is 30 KHz. [4]
 - c) Explain OFDM transmitter in brief. [4]
9. a) Briefly Explain Broadcast Control Channel of GSM system. [5]
 - b) What is a significance of spectrum Management? What are the functions of the regulatory? [2+3]

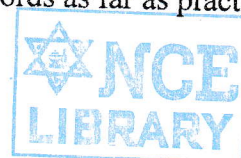
Erlang B Traffic Table												
Maximum Offered Load Versus B and N												
B is in %												
N/B	0.01	0.05	0.1	0.5	1.0	2	5	10	15	20	30	40
16	5.339	6.250	6.722	8.100	8.875	9.828	11.54	13.50	15.18	16.81	20.30	24.54
17	5.911	6.878	7.378	8.834	9.652	10.66	12.46	14.52	16.29	18.01	21.70	26.19
18	6.496	7.519	8.046	9.578	10.44	11.49	13.39	15.55	17.41	19.22	23.10	27.84
19	7.093	8.170	8.724	10.33	11.23	12.33	14.32	16.58	18.53	20.42	24.51	29.50
20	7.701	8.831	9.412	11.09	12.03	13.18	15.25	17.61	19.65	21.64	25.92	31.15

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1. Discuss the features introduced from second to fifth generations of cellular communication system. [4]
2. Explain handover process in cellular system. Write about various practical handover consideration. [4+4]
3. a) What do you mean by fading? Explain various factors influencing small-scale fading. Derive the relation for Doppler's shift. [1+2+5]
- b) Assume free space propagation, a receiver is located 10 km away from a 50W transmitter. The carrier frequency is 900 MHz, antenna gain at transmitter and receiver is 1 and 2 respectively. Find:
 - (i) The power received at the receiver. [2]
 - (ii) The magnitude of the E-field at the receiver. [2]
 - (iii) The rms voltage applied to the receiver input. [2]
 - (iv) The power flux density. [2]
4. Explain with block diagram the concept of orthogonal frequency-division multiplexing (OFDM). [8]
5. a) What is training and tracking mode in an adaptive equalizer? [4]
- b) Compare between feedback and maximum ratio combining technique used in antenna diversity. [4]
6. Why we need speech coding techniques? Explain the basic concept of VOCODER. [2+6]
7. a) Compare between FDMA and TDMA. Explain the operation of FHMA. [4+4]
- b) If total bandwidth is 12.5 MHz, guard bandwidth is 10 KHz and channel bandwidth is 30 KHz, find the number of channels available in an FDMA system. [4]
8. a) Explain the working of all traffic and control channels used in GSM. [8]
- b) Write short notes on: [2×4]
 - (i) WiMAX
 - (ii) Forward CDMA channel

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1. What are the different features of 4G other than 3G mobile communication system? [5]
2. A mobile radio system where each user averages three calls per hour and each call lasting an average of 5 minutes.
 - a) What is traffic intensity for each user?
 - b) Find the number of users that could use the system with 1% blocking if only one channel is available.
 - c) Find the number of users that could use the system with 1% blocking if five trunked channels are available.
 - d) If the number of users you found in (c) is suddenly doubled, what is the new blocking probability of the five channel trunked mobile radio system? Justify whether the performance is acceptable or not. [2+2+2+3]

No. of Trunks (N)	Traffic (A) in erlangs for B%							
	0.1%	0.2%	0.5%	1%	1.2%	3%	5%	7%
1	0.001	0.002	0.005	0.010	0.012	0.031	0.053	0.075
2	0.046	0.065	0.105	0.153	0.168	0.282	0.381	0.470
3	0.194	0.249	0.349	0.455	0.489	0.715	0.899	1.06
4	0.439	0.535	0.701	0.869	0.922	1.26	1.52	1.75
5	0.762	0.900	1.13	1.36	1.43	1.88	2.22	2.50

3. Given a cordless phone operating at 52000 kHz frequency which has a range of 50m. Determine the free space path loss. [6]
4. a) Explain log-distance path loss model. [4]
 - b) Employing the Okumura model compute the transmitter and receiver separation distance if median loss is 167 dB when the carrier frequency is 2.1 GHz. Assume height of transmitting antenna is 40 m, height of receiving antenna is 2 m, for a large city. [Amu = 34 dB, Garea = 0 dB] [8]
5. Explain the significance of Coherence Bandwidth and Coherence Time in mobile radio propagation. [4]
6. Why we need M-ary QAM? Compare 16-QAM and 16-PSK using its constellation diagram. [2+6]
7. What do you understand by RAKE Receiver? Explain the working of M branch RAKE receiver. [3+5]
8. Discuss the different characteristics of speech signal. [6]
9. Draw and explain the frame structure for GSM. Explain the working of all traffic and control channels used in GSM. [4+6]
10. a) What are the major disadvantages of spread spectrum system? Explain the implementation of CDMA system with appropriate example. [2+6]
 - b) Calculate the number of radio channels available in the FDMA system with following data. A US analog mobile phone system is allocated 12.8 MHz for every simplex band. The total spectrum allocated is 12.8 MHz, the guard bandwidth is 10 KHz and the channel bandwidth is 30 KHz. [4]

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1. Briefly describe the evolution of wireless communication from 1G to 4G in terms of technology advancement. [6]
2. Compare and contrast between improper handoff situation and proper handoff situation. Explain cellular concept for $N=7$. [4+3]
3. a) What are the advantages of two ray propagation model over free space path loss model? Derive the equation of path loss using two ray model with appropriate diagram. [2+6]
- b) Estimate the appropriate distance that should be maintained for reverse link between one BTS and mobile with appropriate link budget diagram.
 - i) Mobile is connected to antenna with 20dBi gain, with a transmitting power of 15 dBm and a receive sensitivity of -75dBm.
 - ii) BTS is connected to antenna with 5 dBi gain, with a transmitting power of 20dBm and a receive sensitivity of -80dBm.
 - iii) Cables in both systems are short, with a loss of 3-dB at each side at 900 MHz frequency of operation.

Use Okamura's model for mean path loss where $G(\text{Area}) = 9\text{dB}$, $A_{\text{mu}}=43\text{ dB}$, $h_{\text{re}}=10\text{m}$ and $h_{\text{te}}=100\text{m}$.

(Link margin (reverse link) = 10 dB). [10]
4. What are MSK and GMSK modulation techniques? What are the advantages of spread spectrum modulation technique in wireless communication? [4+4]
5. Explain how diversity improves the quality of network. Explain the operation of RAKE receiver with appropriate block diagram. [2+4]
6. What are different characteristics of speech signal? Explain linear predictive coder with block diagram. [3+5]
7. Explain the term Multiple Access? What do you mean by near for effect in CDMA and state all possible near for effect mitigation techniques. [5+4]
8. Explain the frame structure of GSM. Differentiate between GSM and CDMA standards. [5+3]
9. Write short notes on: (*Any Two*) [5×2]
 - a) Wifi
 - b) Regulatory issues
 - c) Frequency Hopping Multiple Access (FHMA)

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INSTITUTE OF ENGINEERING
Examination Control Division
2075 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Explain briefly the evolution of different generations of cellular systems. [4]
2. Define handoff margin with appropriate figure. [3]
3. A telephone network company needs to expand its capacity based on demand on a city. A group of engineer was selected to find the solution. Among the solution sectoring and cell splitting were major technique for expansion purpose. Being an cellular planning engineer which option do you think is best and why? [5]
4. What is "small scale fading"? Describe briefly its types in radio propagation. Explain the factors which influence small scale fading. [2+4+4]
5. Determine the smallest symbol period T_s , and thus the greatest symbol rate that must be sent through RF channel with given power delay profile without using an equalizer.

Power [dB]	0	0	-10	-20
Delay [us]	0	50	75	100

- Modulation provides suitable BER performance whenever $\sigma_\tau/T_s \leq 0.1$. [8]
6. Explain any two outdoor propagation models used in mobile radio environment. [3+3]
 7. Explain the transmission and detection process of QPSK modulation technique. [6]
 8. Describe a signal processing operation that minimizes the effects of ISI. Explain various space diversity techniques. [2+6]
 9. What are the characteristics of spread signal? Explain the operation of linear predictive coder. [2+6]
 10. What is space division Multiple Access? Explain any two hybrid spread spectrum multiple access technique which minimizes the effect of near for effect. [2+6]
 11. Briefly explain different traffic and control channels used in GSM? [8]
 12. Write short notes on: (Any two) [2×3]
 - a) Frequency Hopped Multiple Access
 - b) Viterbi decoding algorithm
 - c) Wireless local Area Network (WLAN)

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV/ II	Time	3 hrs.

Subject: - Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Compare and contrast the first, second, third and forth generation of mobile communication standards in terms of technology advancement. [6]
2. a. Why does minimizing reuse distance maximize spectral efficiency of a cellular system? [4]
b. For a seven cell reuse pattern, find the minimum distance between centers of co-channel cells. Area of each cell is uniform and is equal to 23 square km. [4]
3. Estimate the feasibility of a 10-km wireless link in suburban area, with one access point and one client radio, using Okumura model for path loss. The median attenuation value is 20 dB and gain due to environment is 13 dB. The height of access point antenna is 100 m and that of client antennal is 10 m: [12]
 - a. Access point is connected to antenna with 5-dBi gain, with a transmitting power of 20-dBm and a receive sensitivity of -80-dBm
 - b. Client is connected to antenna with 20-dBi gain, with a transmitting power of 15-dBm and a receive sensitivity of -75-dBm
 - c. Cables in both systems are short, with a loss of 3-dB at each side at 2.4-GHz frequency of operation.
4. What is known as scattering? Derive an expression for two ray ground reflected model. [2+8]
5. Explain the operation of OFDM with an appropriate block diagram. [8]
6. Why is there a need to implement diversity? Describe the various diversity combining techniques. [4+6]
7. Describe the operation of any two source coders used in speech coding. [6]
8. Explain the principle of Frequency Hopping Multiple Access. Briefly describe two hybrid spectrum multiple access technique which can mitigate near-far problem. [4+6]
9. Write short notes on any two: [5+5]
 - a. Specifications of GSM.
 - b. Regulatory issues
 - c. Convolutional encoding and decoding

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Briefly explain development of mobile communication and evolution path upto 3G technology. [4]
2. (a) Explain handover process in cellular system. Mention various types of handover with application [6]
- (b) Obtain the expression $\frac{S}{I} = \frac{n\sqrt{3N}}{i_0}$ where symbols have their usual meaning. [4]
3. Explain Okumura model of outdoor radio propagation. Determine the median path loss for T-R separation of 50Km, transmit antenna effective height of 100m and receive antenna effective height of 10m in a suburban area with correction factor of 9dB at 900MHz. Assume median attenuation relative to free space 43dB. [4+4]
4. Explain Diversity techniques used in wireless communication. Give brief description of combining methods used for Space Diversity. [4+4]
5. Describe Spread Spectrum Multiple Access, SSMA variants and application with suitable diagram. [8]
6. Describe Waveform and Voice Coding techniques. Mention characteristics of speech. Draw a suitable diagram of GSM CODEC. [4+2+2]
7. Explain Doppler Spread and Coherence Bandwidth. Classify fading on the basis of RMS Delay spread and Coherence Time. [4+4]
8. Explain briefly channel structure of GSM. Show that TDMA frame efficiency cannot reach 100% in GSM [6+2]
9. Compare system architecture of CDMA with LTE. Mention function of entities in the architecture. [3+5]
10. Write Short Note (Any Two) [2X5]
 - a. Adaptive Equalization
 - b. QPSK
 - c. Turbo Coding

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Digital Signal Processing (EX 753)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Explain the general application areas of digital Signal Processing. Consider a continuous time signal $x(t) = \sin 2000\pi t + 5\cos 12000\pi t + 10\sin 6000\pi t$.

a) What is the discrete time signal obtained after sampling the signal at a sampling rate of 5000 samples per second for $0 \leq n \leq 3$? [3+4]

2. Define causal and stable system with examples. [3]

3. Find a convolution between two signals $x[n]$ and $h[n]$ where,

$$x[n] = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad h[n] = \begin{cases} a^n & 0 \leq n \leq 6 \\ 0 & \text{otherwise} \end{cases} \quad (a > 1). \quad [8]$$

4. Using long division determine the inverse Z-transform of,

$$X(z) = \frac{1}{1 - \left(\frac{3}{2}\right)z^{-1} + \left(\frac{1}{2}\right)z^{-2}} \quad \text{when, ROC: } |z| > 1 \quad \text{and ROC: } |z| < \frac{1}{2} \quad [6]$$

5. Define ROC. Explain the properties of ROC with suitable examples. [1+4]

6. Why we need a DFT? Find 8-point DFT of sequence $x[n] = \{1, 0, 2, 0, -1, 1, 1\}$ using Decimation in Time Fast Fourier Transform (DITFFT) algorithm. [1+6]

7. State Multiplication of two DFTs property of DFT. Find $x_3[n]$ if DFT of $x_3[n]$ is given by $X_3(k) = X_1(k)X_2(k)$ where $X_1(k)$ and $X_2(k)$ are 5-point DFT of $x_1[n] = \{1, 2, 3, -1, 5\}$ and $x_2[n] = \{2, 1, -3\}$ respectively. [1+5]

8. For a system with poles at $0.45 \pm j1.06$ and zero at $0.58 \pm j2.06$. Plot the location of poles and zeroes in the z-plane and also plot the magnitude response of the system. [8]

9. Convert the following filter into a lattice ladder structure.

$$H(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3} \quad [6]$$

10. Design a low pass digital filter by Bilinear Transformation method to an approximate Butterworth filter, if passband edge frequency is 0.25π radians and maximum deviation of 0.99 dB below 0 dB gain in the passband. The maximum gain of -14.85 dB and frequency is 0.59π radians in stopband, Consider sampling frequency 0.5 Hz. [11]

11. List out the key points of windowing and design the symmetric FIR low pass filter for which desired frequency response is expressed as

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & \text{for } |\omega| \leq \omega_c \\ 0 & \text{elsewhere} \end{cases}$$

The length of the filter should be 7 and $\omega_c = 1$ radians/sample. Use Hanning window as a prototype.

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Wireless Communication (EX 751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

***Hata Model**

$$L_{50} (\text{urban}) (\text{dB}) = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$$

For medium sized city

$$a(h_{re}) = (1.1 \log f_c - 0.7) h_{re} - (1.56 \log f_c - 0.8) \text{ dB}$$

For large city

$$a(h_{re}) = 8.29 (\log 1.54 h_{re})^2 - 1.1 \text{ dB for } f_c \leq 300 \text{ MHz}$$

$$a(h_{re}) = 3.2 (\log 1.75 h_{re})^2 - 4.97 \text{ dB for } f_c \geq 300 \text{ MHz}$$

$$L_{50} (\text{Suburban}) (\text{dB}) = L_{50} (\text{urban}) - 2 [\log (f_c / 28)]^2 - 5.4$$

$$L_{50} (\text{rural}) (\text{dB}) = L_{50} (\text{urban}) - 4.78 (\log f_c)^2 + 18.33 \log f_c - 40.94$$

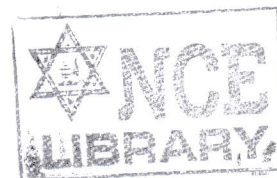
1. a) Compare various generations of mobile communications up to the fourth generation. [4]
b) What are various practical handoff considerations? Explain. [4]
2. Determine: [6]
 - a) the cell cluster size
 - b) the number of cell clusters in the service area
 - c) the maximum number of users in service area at any instant
3. a) Discuss what is meant by the term FADING. Describe briefly its types in radio propagation. [2+2]
b) Let us consider a medium sized city and assume the typical GSM downlink parameters. The Base Station (BS) is transmitting with power 50W. The minimum acceptable received power at Mobile Station (MS) is -91 dBm. The carrier frequency is 900 MHz, the height of BS is 30m and height of MS is 1m. Estimate the maximum cell radius and corresponding cell area using Hata Model. (*The expression below should be provided in the question) [6]
4. a) Describe Direct Sequence and Frequency Hopped Spread Spectrum Techniques. [4]
b) State the advantages and disadvantages (two of each): [4]
 - (i) Frequency Division Multiple Access (FDMA)
 - (ii) Time Division Multiple Access (TDMA)
 - (iii) Code Division Multiple Access (CDMA)
- c) Define the terms Coherence Bandwidth and Coherence Time explaining their significance in mobile radio propagation. [4]
5. a) Explain briefly adaptive equalization algorithms (any two) [4]
b) Explain various space diversity techniques along with block diagrams. [4]
6. a) Describe Outdoor Propagation Models (any two) [5]
b) Describe vocoders with block diagram. Briefly explain different kind of vocoders. [2+3]
7. a) Explain with necessary diagram system architecture of GSM. Mention functions of various Blocks in the architecture. [7]
b) What is channelization code? Explain briefly forward channels in cdma IS-95. [4]
8. Write short notes: (Any three) [3×5]
 - a) WiMAX
 - b) Handover

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Differentiate between 2G and 3G with examples of appropriate technologies used. Explain the terms prioritized handoff and cell dragging. [4+2]
2. State the difference between large scale and small scale propagation model. Explain the different propagation mechanisms which have impact on propagation in mobile environment. [3+6]
3. A BS transmitter has a power output of 10 watts operating at a frequency of 250 MHz. The transmitter is connected by 20 m of an RF coaxial cable, which has a loss of 3-dB/100 m specification, to an antenna that has a gain of 9dBi. The receiving antenna is 25 km away and has a gain of 4 dBi. There is negligible loss in the receiver feeder line, but the receiver is mismatched; the receiving antenna and feeder cable are designed for 50 ohm impedance. The receiver impedance loss due to mismatch is of about 0.2 dB. Calculate the power delivered to the receiver, assuming free-space propagation. [8]
4. What do you understand by RACK receiver? Explain the working of a M branch RACK receiver. [8]
5. What are the different characteristics of speech signals? How they are used in designing of coders? [8]
6. What is self jamming problem in CDMA? Explain the operation of FHMA with the help of block diagram. Explain any two hybrid spread spectrum multiple access technique along with their advantage and disadvantage. [2+4+6]
7. Explain the working of all traffic and control channels used in GSM. [8]
8. Explain with block diagram the concept of Maximum Likelihood Sequence Estimation. Define time diversity. Explain two implementations of time diversity. [6+1+5]
9. Write short notes on: [3×3]
 - a) WiMax
 - b) LTE
 - c) Viterbi decoding algorithm



Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Briefly describe the evolution of wireless communications from first to third generation. [6]
2. A city with a coverage area of 1500 sq km is covered with a 12-cell system each with a radius of 1.387 km. If the total spectrum allocated is 28.5 MHz with a full duplex channel bandwidth of 25 MHz. Assume a GOS of 0.02 for a blocked calls cleared system, is specified and the offered traffic per user is 0.03 Erlangs and traffic intensity of each cell is 84 Erlang, compute: [12]
 - (a) the number of cells in the service area
 - (b) the number of channels per cell
 - (c) the maximum carrier traffic
 - (d) the total number of users that can be served for 2% GOS
 - (e) the number of mobiles per unique channel.
 - (f) Theoretical maximum number of users that could be served at one time by the system.
3. Explain the transmitter and receiver of DPSK modulation scheme. Briefly explain about pseudo-noise (PN) sequence. Why is it used? [4+2+2]
4. What do you understand by frequency reuse concept? Define Co-channel reuse Ratio in details. [6]
5. Explain the mobile radio propagation in terms of large scale path loss and small scale fading. [8]
6. Explain the Transmission and Detection of QPSK modulation technique. [8]
7. Why we need equalization technique in communication? Explain the basic equalization technique. [2+5]
8. What are the characteristics of speech signal? Explain the operation of Linear predictive coder. [2+6]
9. Explain the different types of spread spectrum multiple access techniques. Compare FDMA with CDMA. [6+2]
10. What is small scale fading? Describe the different factor that influences the small scale fading. [3+6]



Exam.	Regular / Back		
	BE	Full Marks	80.
Level	BE	Pass Marks	32
Programme	BEX	Time	3 hrs.
Year / Part	IV / II		

Subject: - Wireless Communication-(EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- ① Explain the evolution of cellular radio 1G to 3G. [4]
2. Prove that for a hexagonal geometry the co-channel reuse ratio is given by $Q = \sqrt{3}N$; Where $N = i^2 + j^2 + ij$. A cellular service provider decides to use a digital TDMA scheme which can tolerate a Signal-to-Interference Ratio of 15 dB in the worst case. Find the optimal value of N for [4+4]
 - a) Omni directional antennas
 - b) 120° Sectoring
 - c) 60° Sectoring

[Use path loss exponent of 4 and consider trunking efficiency]
3. Derive the expression for phase difference in two ray free space propagation model. [8]
4. A mobile is located 5 km away from a base station and a vertical $\lambda/4$ monopole antenna with a gain of 2.55 dB to receive cellular radio signals. The electric field at 1 km from the transmitter is measured to be 10^{-3} V/m. The carrier frequency used for this system is 900 MHz. [6]
 - a) Find the length and effective aperture of the receiving antenna.
 - b) Find the received power at the mobile using two ray ground reflection model assuming the height of the transmitting antenna is 50 m and the receiving antenna is 1.5 m above ground.
5. What is the difference between path loss and fading of signal? Explain time dispersion fading and its types. (2+6)
6. Explain the transmitter and receiver of OQPSK modulation. Discuss why $\pi/4$ -QPSK is more preferred than OQPSK modulation. [5+2]
- ⑦ Why diversity is important in wireless communication system? Explain different types of diversity techniques. [2+6]
- ⑧ Explain the operation of formant vocoder. What are the characteristics of speech signal? [4+4]
- ⑨ Explain the terms Multiple access, Time Division CDMA (TCDMA) and Time Division Frequency Hopping as related to wireless communication system. [7]
- ⑩ What is a multiple access technique? Explain TDMA, CDMA and SDMA. [2+6]
- ⑪ What are the basic signal processing operations to be performed to convert a speech signal into a radio signal and back in GSM? Describe briefly. [8]

Subject: - Wireless Communication (EX751)

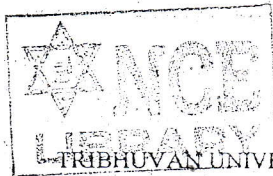
- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Discuss the evolution from 1G to 2G, 2.5G in the case of cellular network based on TDMA. [4]
2. Describe the techniques used for enhancing the capacity and coverage in cellular radio network. [8]
3. a) With appropriate expressions, distinguish between Rayleigh fading channel and Rician fading channel. [2]
- b) A wireless channel is characterized by the following power-delay profile: [6]

Power [dB]	0	-10	-20	-23
Delays [ns]	0	100	200	400

Determine the root mean square (rms) delay spread and the 90% coherence bandwidth of the above channel. Is this channel flat fading or frequency selective fading for:

- i) An AMPS system with transmission bandwidth 30 kHz?
- ii) A GSM system with transmission bandwidth 200 kHz?
4. Explain any two outdoor propagation models used in mobile network environment. [3+3]
5. What are the parameters of mobile multipath channel? Explain. [7]
6. What is an OFDM? Generalize the modulation and demodulation technique of OFDM. [8]
7. a) Discuss and compare different types of antenna diversity technique. [4]
- b) Explain with block diagram the concept of Maximum Likelihood Sequence Estimation equalization. [4]
8. What is a channel coding? Explain types of linear predictive coder. [2+6]
9. a) Define near-far effect. Briefly describe any one hybrid spread spectrum multiple access technique which can mitigate the near-far problem. [2+2]
- b) What are the advantages of TDMA cellular system over FDMA cellular system? [4]
10. Explain the principle of FHMA. What do you mean by near-far effect in CDMA? How is it solved? Explain. [3+4]
11. What is GSM and CDMA standard? Explain the architecture of GSM. [4+4]



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TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING**Examination Control Division**
2070 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject - Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

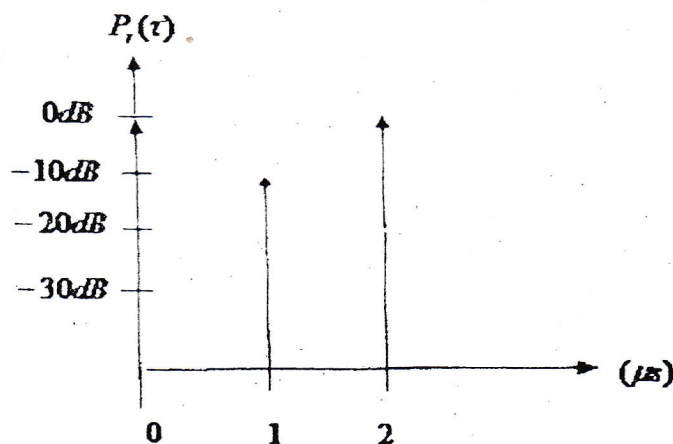
1. List the significant improvements introduced in the second, third and beyond third generation standards of cellular communication systems. [6]
2. Explain the difference between co-channel and adjacent channel interference. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + ij + j^2$ is the cluster size. If 20 MHz of total spectrum is allocated for a duplex (i.e. bidirectional) wireless cellular system and each simplex (i.e. one-way) channel has 25 KHz of bandwidth, find [3+4+3]
 - a) The number of duplex channels, and
 - b) The total number of channels per cell, assuming a cluster size of $N = 4$.
3. Explain indoor propagation models (any two). [8]
4. Determine the radio coverage range of a base station that transmits a RF signal at 150 W, given the receiver threshold level is -104 dBm. Assume that the path loss at the first meter is 15 dB in a mobile radio propagation condition. (Path loss exponent = 4) [6]
5. Discuss the principle of Orthogonal Frequency Division Multiplexing modulation scheme. Briefly explain different types of spread spectrum modulation techniques. [4+4]
6. What is diversity? Explain any two types of diversity techniques in detail. [2+6]
7. What is vocoder? Explain any two predictive coders. [2+6]
8. Define multiple access. What are the merits and demerits of Code Division Multiple Access? If a normal GSM time slot consists of 6 trailing bits, 8.25 guard bits, 26 training bits, and 2 traffic bursts of 58 bits of data, find the frame efficiency. [2+6+4]
9. Draw and explain the frame structure for GSM. Describe how various traffic and control channels are used while making a call in GSM system. [4+4]
10. Write short notes on: (any two) [6]
 - a) Viterbi Decoding Algorithm
 - b) Doppler Spread and Coherence Time
 - c) GMSK Modulation Technique

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: Wireless Communication (EX751)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Explain the evolution of wireless communication in terms of technology and worldwide market penetration. [6]
2. What is hand off? Explain its strategy used in GSM. [8]
3. Determine the propagation path loss for signal at 800 MHz, with a transmitting antenna height of 30 m and a receiving antenna height of 2 m, over a distance of 10 km, using two-ray mobile point-to-point propagation model. How is it compared with that of free-space propagation path loss model? [4+4]
4. Define Doppler spread. Describe the types of small scale fading based on Doppler spread. Calculate the mean excess delay and rms delay spread for the multipath profile given below. Estimate the 90% and 50% coherence bandwidth of the channel. [4+4]



5. What are MSK and GMSK modulation techniques? Draw the block diagram of OFDM modulator and demodulator and explain them. [8]
6. Describe the fundamentals of equalization with respect to communication system? Explain with block diagram the function of Rake receiver. [4+4]
7. Why we need speech coding techniques? Explain the basic concept of VOCODER. [4+4]
8. What is multiple Access technique? Compare FDMA with CDMA. [2+4]
9. Draw the architecture of GSM and explain it. [8]
10. Write short notes on: [4×3]
 - a) Rayleigh and Ricean fading distribution
 - b) Regulatory issues in wireless systems
 - c) Viterbi decoding algorithm